

Related-Work

Criteria	Paper 1	Paper 2	Paper 3	Paper 4	Paper 5
Title	Gesture recognition from surface electromyography signals based on the SE-DenseNet network	Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification	Hybrid Convolutional Vision Transformer for Robust Low-Channel sEMG Hand Gesture Recognition : A Comparative Study with CNNs	A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals	Surface EMG-Based Hand Gesture Recognition Using a Hybrid Multistream Deep Learning Architecture
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Year	2025	2024	2025	2026	2026
Dataset	NinaPro DB2 & DB4	Public multi-channel sEMG benchmarks	NinaPro DB3 (Amputees) & Compact	8-channel sEMG Myo Armband Array (8	FORS-EMG Dataset (Multi-gesture

		(8–12 channels)	Myo (8 channels)	healthy subjects)	surface EMG)
Application Area	Prosthetic Control & Physical Rehabilitation	Human-Computer Interaction (HCI) & Orthopedics	Low-Channel Prosthetic Control & Wearable Devices	Real-time Assistive Prosthetics , AR/VR Control	High-Precision Biometric Gesture Identification
Method / Model	SE-DenseNet (DenseNet-101 backbone + Squeeze-and-Excitation attention mechanism)	Spatiotemporal Graph Neural Network (ST-GCN)	Convolutional Vision Transformer (CViT)	Fast sEMG Graph Neural Network (sEMG-GNN) based on Pearson correlation activation graphs	Multistream Hybrid TCN + Bi-LSTM/GRU + Transformer + ArcFace Loss
Type of GNN	N/A (Non-GNN: CNN with Channel Attention)	Spatiotemporal Graph Convolutional Network (ST-GCN / Physical Topology GNN)	(Non-GNN: Hybrid CNN + Self-Attention)	Dynamic Functional Correlation Graph Neural Network (GNN / GCN)	(Non-GNN : Hybrid TCN-RNN-Transformer)

Metrics	Accuracy: 85.93% (DB2), 82.39% (DB4)	Superior classification accuracy over standard 2D-CNNs; high robustness under reduced node conditions	Accuracy: 94.2% on NinaPro DB3 amputees, 96.6% on Myo armband	Accuracy: 99.0%; Execution Latency: 48 ms (M1 Pro CPU)	Within-subject Accuracy: 96.4%; Cross-subject Accuracy: 74.0%
Strengths	Enhances feature reuse; effectively mitigates gradient vanishing while focusing on key channel features.	Models physical forearm electrode geometry as graph nodes for explicit spatial aggregation.	Superior robustness across amputee and non-amputee subjects; captures long-range temporal dependencies.	Ultra-low computational budget; constructs dynamic graphs reflecting functional muscle activation correlations.	ArcFace angular margin loss creates highly compact intra-class features and distinct inter-class decision boundaries.
Limitations	High parameter count; high computational overhead; prone to vanishing gradients on deep	Relies on static/fixed physical electrode graph configurations; sensitive to skin shift and sensor displacement.	Self-attention mechanism introduces heavy latency and parameter overhead unsuitable for low-power edge	Tested exclusively on intact, homogeneous subjects; lacks cross-subject and amputee	Massive accuracy drop (~22.4%) under cross-subject evaluation due to severe biological

	inaccurate layers.		microcontrollers.	dataset validation.	domain shift.
Research Gap	Struggles to capture spatial topology and pairwise electrode dynamic interactions in low-density arrays.	Lacks dynamic edge re-weighting to model temporal shifts in muscle co-activation patterns.	High performance relies on self-attention matrix computation rather than lightweight graph structural connectivity.	Missing validation on complex non-stationary sEMG signals (e.g., limb position changes, muscle fatigue, amputees).	Severe sensitivity to domain shift without explicitly modeling generalized graph topological embeddings across subjects.
Relation to this Work	Serves as a baseline benchmark for CNN-based spatial feature extraction without graph-based structural modeling.	Provides the foundation for representing electrode locations as graph nodes, which our work extends with dynamic graph learning.	Highlights the need for high-accuracy gesture decoding in amputee datasets, driving our focus toward lightweight real-time architectures.	Direct inspiration for real-time edge processing ; our model builds on this by addressing domain shift and heterogeneous subject evaluation.	Identifies intra-class/ inter-class feature separation techniques , guiding our loss function design to improve cross-subject generalizability.

Reference

1. **SE-DenseNet (2025):** *Gesture recognition from surface electromyography signals based on the SE-DenseNet network*, ResearchGate / Ninapro DB2 & DB4 Benchmark.
https://www.researchgate.net/publication/388487726_Gesture_recognition_from_surface_electromyography_signals_based_on_the_SE-DenseNet_network
2. **ST-GCN (2024):** *Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification*, IEEE TNSRE, 2024.
<https://ieeexplore.ieee.org/document/10818442/>
3. **CViT (2025):** *Hybrid Convolutional Vision Transformer for Robust Low-Channel sEMG Hand Gesture Recognition: A Comparative Study with CNNs*, MDPI Sensors, 2025.
<https://www.mdpi.com/2313-7673/10/12/806>
4. **Fast sEMG-GNN (2026):** *A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals*, arXiv:2607.07850, 2026.
<https://arxiv.org/abs/2607.07850>
5. **Multistream TCN-ArcFace (2026):** *Surface EMG-Based Hand Gesture Recognition Using a Hybrid Multistream Deep Learning Architecture*, MDPI Sensors, 2026.
<https://www.mdpi.com/1424-8220/26/7/2281>